Graph Sketching: Parabolas

Questions 1–5

Question 1. Sketch each of the following curves:

a) $y^2 = 12x$

b) $y^2 = 2x$

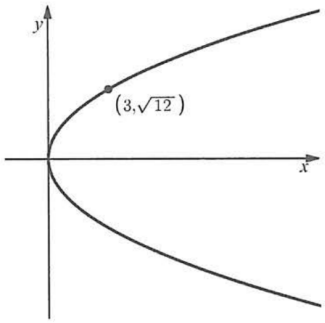
c) $y^2 = -6x$

Question 2. Each of the following curves are of the form

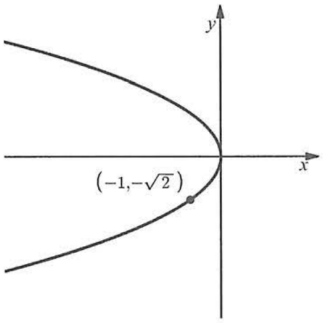
$$y^2 = ax$$

where a is a constant. In each case state the value of a

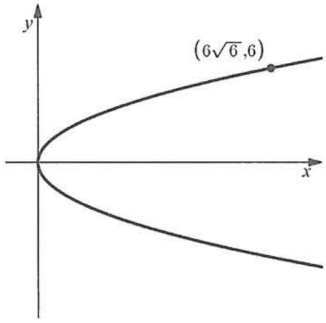
a)



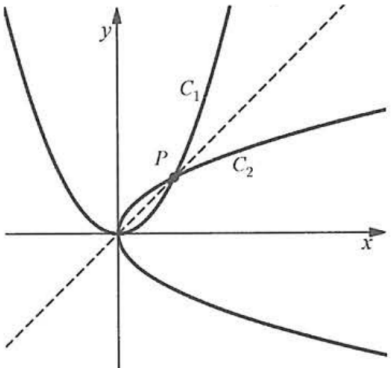
b)



c)



Question 3. A curve C_1 is given by the equation $y = kx^2$ where k is constant. A second curve, C_2, is shown on the diagram which is the image of C_1 under a reflection over the line $y = x$



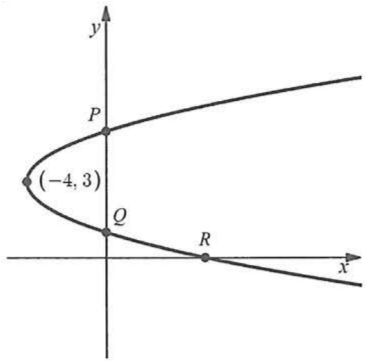
a) If $k = \frac{1}{4}$, find an equation for curve C_2 and the coordinates of the point P
b) If $k = 3$, find an equation for curve C_2 and the coordinates of the point P
c) If $k = a$, find an equation for curve C_2 and the coordinates of the point P

Question 4. State the coordinates of the vertex for each of the parabolas below:
a) $y - 4 = (x - 2)^2$
b) $x - 4 = (y - 2)^2$
c) $y + 1 = (x - 5)^2$
d) $x + 1 = (y + 5)^2$

Question 5. A parabola is given by the equation

$$(y-a)^2 = x - b$$

where a and b are constants is shown in the diagram below. The points of intersection with the y-axis are P and Q and with the x-axis is R



a) State the value of a and b
b) Find the coordinates of points P, Q and R